MC-KELLY TAMUNOTENA PEPPLE

AI Engineer | Machine Learning Researcher | Computer Vision Specialist
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Google Scholar: https://scholar.google.com/citations?user=Cuec4loAAAAJ&hl=en
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ACADEMIC PROFILE

Lecturer and researcher specializing in Control Systems, Artificial Intelligence, and Machine Learning applications. Extensive experience in teaching engineering and computing courses, supervising student projects, and conducting interdisciplinary research. Proven record of peer-reviewed publications and conference presentations. Proficient in MATLAB, Simulink, and Python for computational modeling and simulation.

RESEARCH INTERESTS

Artificial Intelligence; Machine Learning; Control Systems; Computational Modeling; Signal Processing

EDUCATION

PhD (In View), Electrical/Electronic Engineering, University of Nigeria, Nsukka (2021–Present)
M.Sc. Systems and Control, Coventry University, United Kingdom (2013–2014)
B.Tech Computer Engineering, Rivers State University, Port Harcourt (2003–2008)
WASC, Enitonna High School (2000)
NECO, Enitonna High School (2000)
FSLC, Boyle Memorial School (1994)

PROFESSIONAL CERTIFICATE

Registered Engineer (COREN), Council for the Regulation of Engineering in Nigeria (2019)

Python Developer Certificate (FreeCodeCamp)

Intermediate Machine Learning (Kaggle)

ACADEMIC & PROFESSIONAL EXPERIENCE

Lecturer III, Electrical/Electronic Engineering Technology
Federal Polytechnic of Oil and Gas, Bonny (2018–Present)
- Teaching undergraduate engineering and computing courses
- Supervising student research and projects
- Conducting and publishing research
- Participating in curriculum development

Mathematics and Technical Drawing Teacher, Port Harcourt International School (2017–2018)
Information Technology Instructor, Kufman Power & Automation Training Centre (2015–2016)
Computer Instructor, Bonny Vocational Centre (2012–2013)
Computer and Mathematics Teacher, Kings and Queens Secondary School (2011–2012)
Mathematics and Physics Teacher, St. Paul Comprehensive High School (2010–2011)
Computer Instructor (NYSC), Ministry of Information, Ebonyi State (2008–2009)
Industrial Training, Bonny LGA (2005–2006)

PUBLICATIONS

Pepple, M.-K. T. (2025). Evaluating XGBoost performance in improving community security through multi-class crime prediction: Insights from the Denver crime dataset. International Journal of Applied Sciences and Smart Technologies.

Pepple, M.-K. T. (2025). Dense neural networks and local interpretable model-agnostic explanations hybrid model for enhanced interpretability. International Journal of Computer Science and Mathematical Theory.

Pepple, M.-K. T. (2025). Comparative analysis of K-nearest neighbors, Naive Bayes, and decision tree algorithms for credit card fraud detection. World Journal of Innovation and Modern Technology.

Pepple, M.-K. T. (2025). Multicriteria decision analysis for stock selection: A comparative study of ELECTRE III with veto, TOPSIS, and PROMETHEE methods. International Journal of Engineering and Modern Technology, 11(2), 137–146.

Pepple, M.-K. T. (2022). Comparison of GoogleNet convolutional neural network to visual geometry group convolutional neural network. Rivers State University Journal of Biology and Applied Sciences.

Pepple, M.-K. T. (2022). Efficiency of state-dependent proportional integral plus over proportional integral plus in the control of air conditioner with free cooling. Irish International Journal of Engineering and Scientific Studies.

Pepple, M.-K. T. (2022). QoS performance evaluation of video streaming across IPv6. Rivers State University Journal of Biology and Applied Sciences.

Pepple, M.-K. T. (2021). Energy efficiency of model predictive control over proportional integral-plus towards control. International Journal of Scientific Research and Engineering Development.

Pepple, M.-K. T. (2020). Impact of node velocity on TCP throughput of mobile ad-hoc data network (MADNET). European Journal of Engineering Technology.

Pepple, M.-K. T. (2020). Experimental assessment of the impact of velocity on UDP end-to-end delay of mobile unicast data network (MUDNET). International Journal of Computing and Technology.

CONFERENCE PRESENTATIONS

EfficientNet and Swin Transformer Framework with Cross-Attention for Face Recognition, IECEC, University of Nigeria Nsukka (2026)

Noise-Aware Feature with Channel Attention U-Net for Clinical Image Denoising, IECEC, UNN (2026)

Raw Material Processing and Utilization for Sustainable Development, Maritime Conference, Federal Polytechnic Bonny (2024)

Fault Detection in Wind Turbine Blades using AI Transfer Learning, UNN (2024)

Improving Community Safety with XGBoost, Federal Polytechnic Bonny (2024)

Control Systems Optimization in Air Conditioner Systems, Federal Polytechnic Bonny (2022)

Zero Inflation Count Regression Models, IEMSC-21, India (2021)

Markov Switching Regression for Economic Growth, IEMSC-21, India (2021)

Shadow Controlled Door Bell Design, Yaba College of Technology (2021)

ADMINISTRATIVE EXPERIENCE

Head of Department, Electrical/Electronic Engineering Technology (2021)
Chairman, Examination Moderation Committee (2021)
Chairman, Appraisal Committee (2021)
Member, School Appraisal Committee (2021)
Member, Disciplinary Committee (2019)

TECHNICAL SKILLS

Programming: Python, MATLAB
Simulation: Simulink
Areas: Artificial Intelligence, Machine Learning, Control Systems

REFEREES

Engr. Dr. S. L. Braide – Rivers State University
Engr. Prof. H. A. Igoni – Rivers State University
Barr. Waripanye Valentine Pepple – Bonny